

Scientific Forgery Image Detection

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Table of Contents

1. Introduction
2. System Architecture
3. User Interface Design
4. Glossary
5. Versions Table
6. References

1. Introduction

1.1 Purpose

Scientific images play an important role in academic research and published studies. However, some images may be manipulated through techniques such as copy-move forgery, where parts of an image are duplicated and placed elsewhere to falsify results.

The purpose of this system is to detect and analyze copy-move forgery in biomedical and scientific images. Users will be able to upload images through a web interface, and the system will process them to identify potentially duplicated regions. This helps support scientific integrity by assisting researchers and reviewers in identifying manipulated figures.

1.2 Intended Audience

This document is intended for:

- Software developers working on the system
- Course instructors evaluating the project
- Students involved in system design
- Future maintainers of the software

1.3 Overview

The system is a web-based application consisting of a frontend interface and a backend server. Users upload images through the frontend, and the backend processes these images for future detection logic development.

At this stage, the system focuses on:

- Project setup
- Frontend image upload structure
- Backend API structure
- Fraudulent region detection workflow using OpenCV
- Basic results display with highlighted suspicious regions

2. System Architecture

2.1 Overview

The system follows a simple client-server architecture.

2.2 Frontend

The frontend is responsible for user interaction.

Responsibilities:

- Provide image upload interface
- Send image data to backend
- Display backend responses
- Show suspicious regions

Technologies:

- React.js
- HTML/CSS

2.3 Backend

The backend handles incoming requests from the frontend.

Responsibilities:

- Receive uploaded images
- Validate image file format
- Run fraudulent region detection
- Return placeholder or future detection results

Technologies:

- Python
- FastAPI
- Uvicorn
- OpenCV
- Numpy
- Pillow

2.4 Detection Module

The detection module analyzes uploaded images for copy-move forgery.

Responsibilities:

- Detect keypoints in images using ORB
- Match keypoints to identify duplicated regions
- Highlight suspicious regions on the output image
- Return detection results to backend

2.5 Workflow Design

Updated Workflow:

1. User uploads biomedical image through frontend
2. Frontend sends image to backend via POST request
3. Backend receives and validates the image
4. Detection module analyzes the image for duplicated regions
5. Suspicious regions are identified and marked
6. Detection results returned to frontend
7. User views highlighted suspicious regions

3. User Interface Design

3.1 Homepage

The homepage introduces the application and provides navigation to upload functionality.

Elements:

- Project title
- Short description
- Navigation to upload page

3.2 Image Upload Interface

This page allows users to submit images for analysis.

Elements:

- Upload button
- File selection option
- Submit button
- Image preview

3.3 Results Display

- Confirmation of upload
- Detection status message
- Marked image showing highlighted suspicious regions
- Match count indicator

4. Glossary

Term	Definition
API	Application Programming Interface used for communication between frontend and backend
Backend	Server-side system that processes requests
Frontend	User interface of the application

Copy-Move Forgery	Image manipulation where parts of an image are duplicated within the same image
React.js	JavaScript library used for building user interfaces
FastAPI	Python framework used for building APIs
Uvicorn	Server used to run FastAPI applications
OpenCV	Open-source computer vision library used for image analysis
ORB	Oriented FAST and Rotated BRIEF — keypoint detection algorithm
Keypoint	A distinctive location in an image used for matching

5. Versions Table

Version	Date	Description	Authors
1.0	May 8, 2026	Initial Software Design Document created for Snapshot 1	Janelle Rivera, Eric Marroquin, Alexis Flores
1.1	May 10, 2026	Updated frontend implementation progress and UI design	Janelle Rivera, Eric Marroquin, Alexis Flores, Alex Lam
1.2	May 12, 2026	Added fraudulent region detection workflow and frontend/backend integration updates	Janelle Rivera, Eric Marroquin, Alexis Flores, Alex Lam

6. References

- FastAPI Documentation: <https://fastapi.tiangolo.com/>
- React Documentation: <https://react.dev/>
- OpenCV Documentation (future reference): <https://opencv.org/>